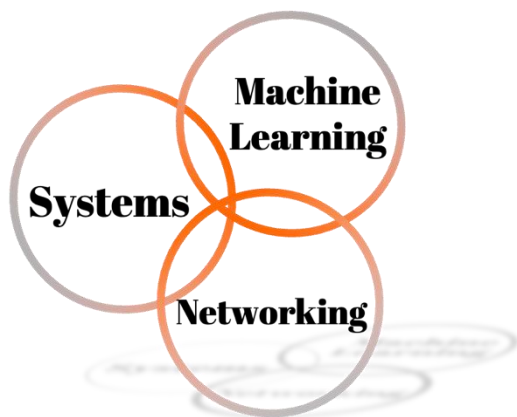
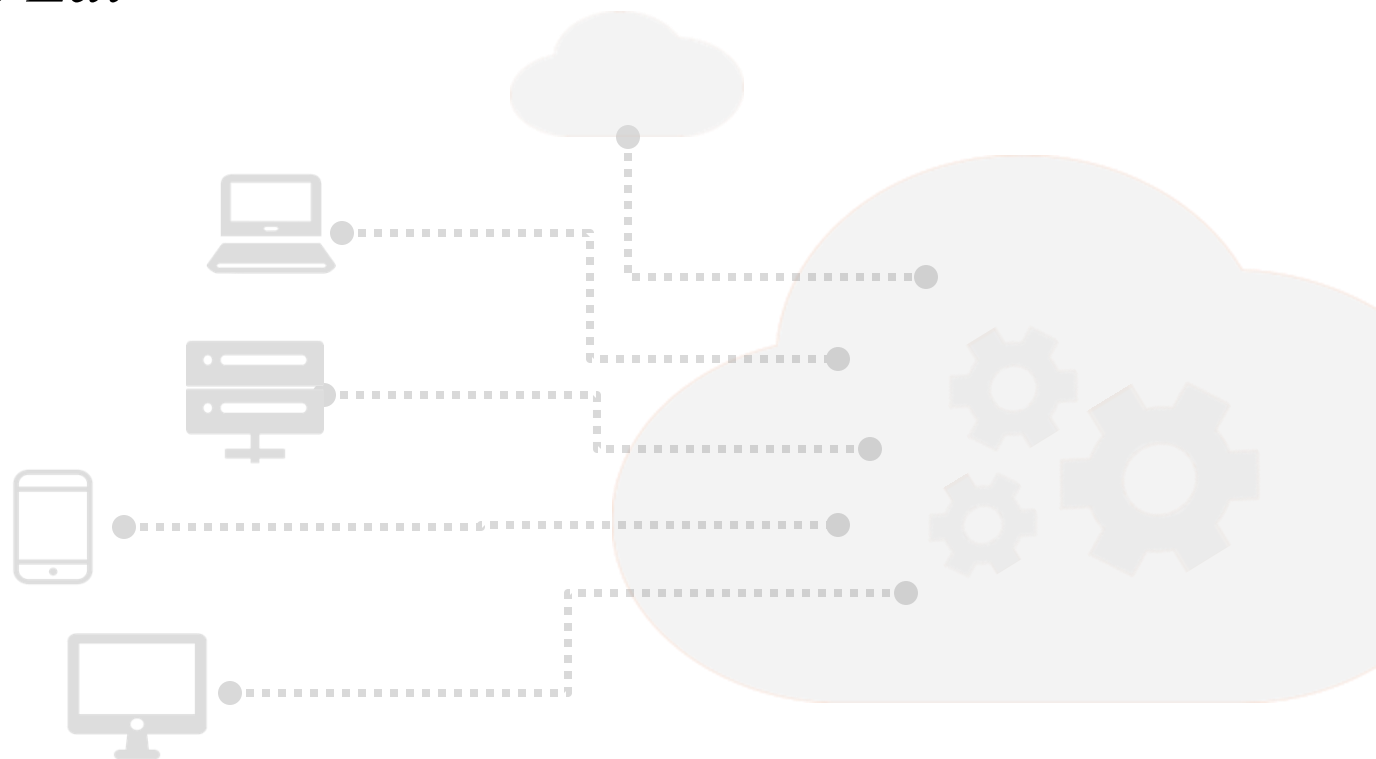




CS 498: Machine Learning Systems

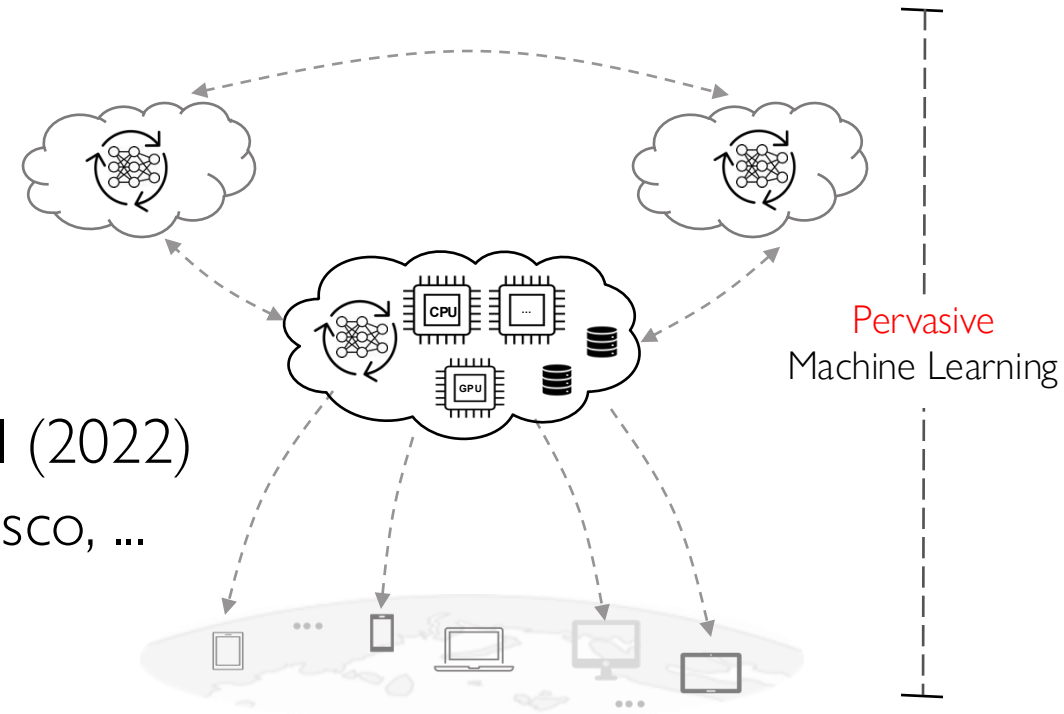
Instructor: Fan Lai

I ILLINOIS

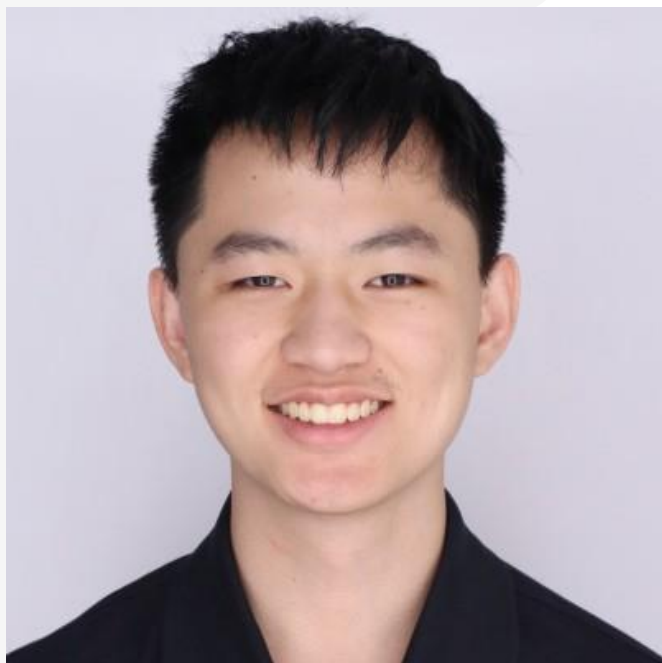


About Instructor (Fan)

- **Assistant Professor in CS**
 - PhD'23: "Minimalist Systems for Pervasive Machine Learning"
- **Research interest**
 - Cloud ML (LLMs, Image/Video Gen)
 - On-device ML, ML4Sys, ML efficiency algs
- **Industry**
 - Visiting Faculty@Google (2023-2024), Meta AI (2022)
 - MLSys adoptions at Google, Meta, LinkedIn, Cisco, ...
- **Office hours:** SC 3128, Wed 3 PM – 4 PM



About Our Two TAs (Jimmy and Jamlee)



Jimmy Shong

(Research: Cloud ML training & inference)



Jianli (Jamlee) Jin

(Research: Edge ML training & inference)

Office Hours: Friday 2-3 PM ([Zoom](#), or in-person for assignment due dates)

Status and Prerequisites

- As of today: ~94 registered
 - If you are not planning to take the class, please drop ASAP
- Systems Courses: OS/databases/distributed systems/networking
 - Equivalent courses are acceptable as well
- ML/AI course is helpful but not required
- Good programming skills (e.g., PyTorch, TensorFlow, ...)
 - Build systems for course project

Course Schedule

- Webpage: <https://github.com/fanlai0990/cs498>
- Discussion, Questions: Please Use Canvas
- In-person lectures, discussion
- Feel free to drop us email too :)

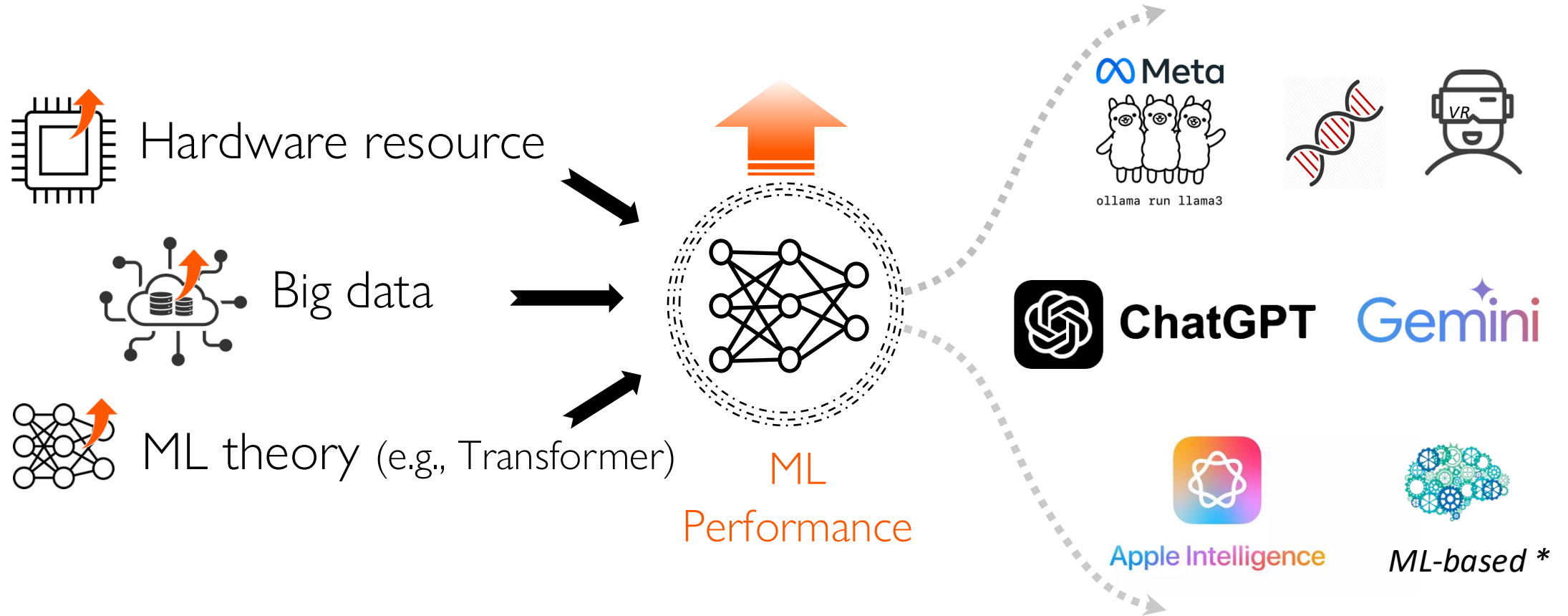
Agenda

- What will you learn in this course?
 - GenAI, systems, networks, ...
- What will you do in this course?
 - Participation, assignment, course project, ...



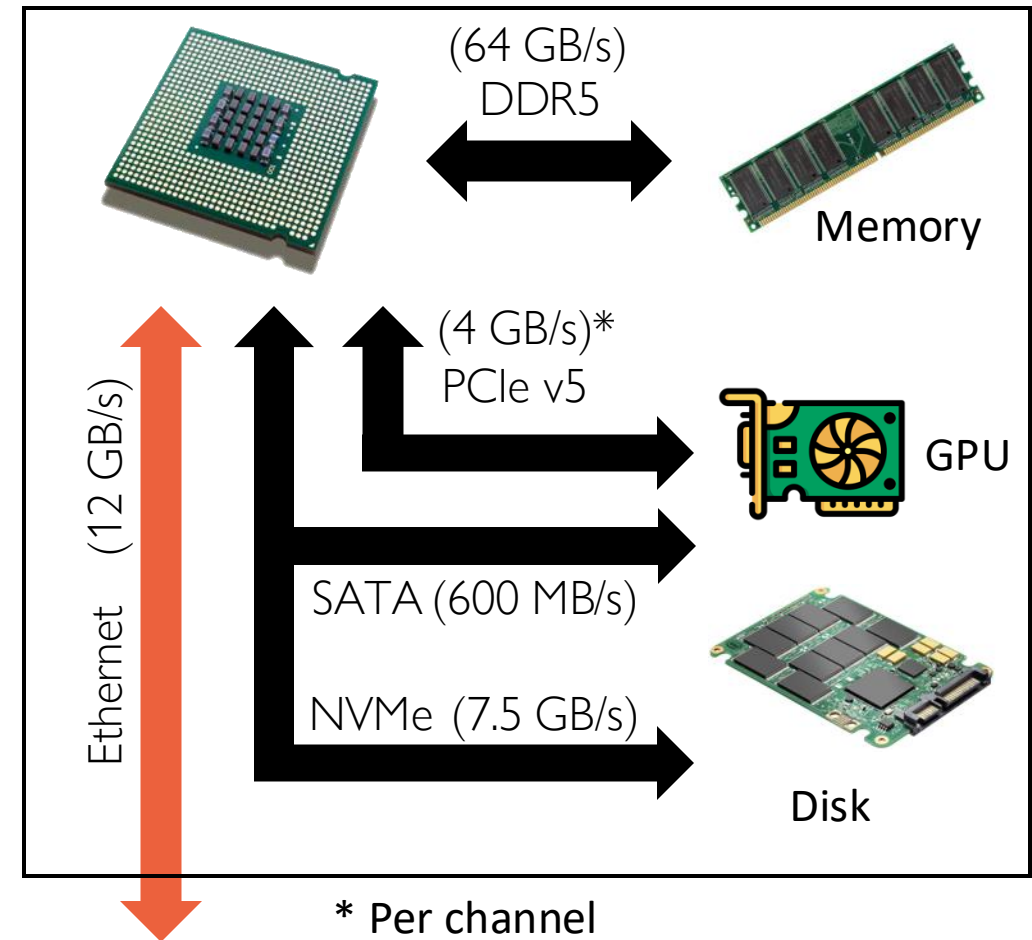
What Do We Talk About When We Talk About “Machine Learning **Systems**”

Cloud Enabled Last Leap in Machine Learning (ML)

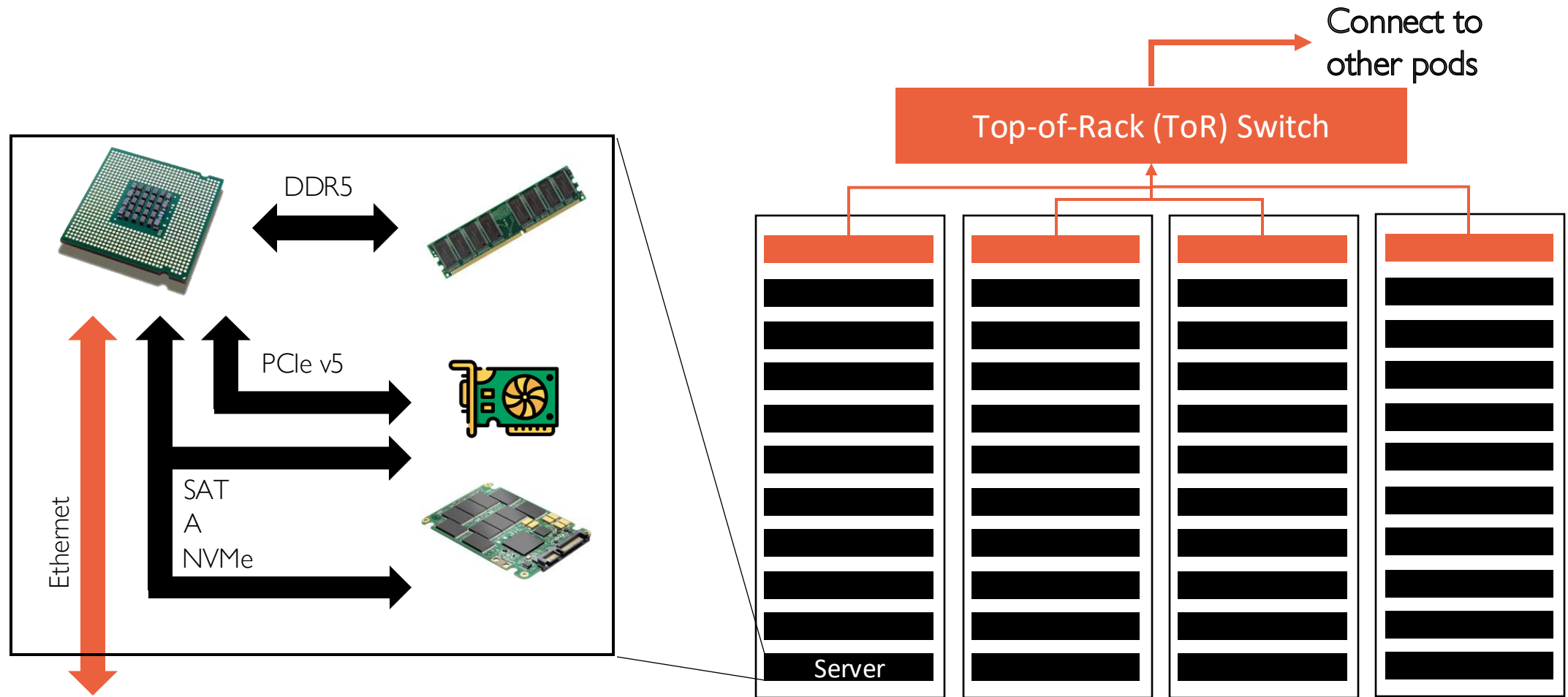


What's in a (Simplified) AI Server?

- **Interconnected compute and storage resources**
 - Different bandwidth and latency constraints
- **Simplified diagram**
 - Doesn't include faster networks such as RDMA, dedicated GPU interconnects such as NVLink

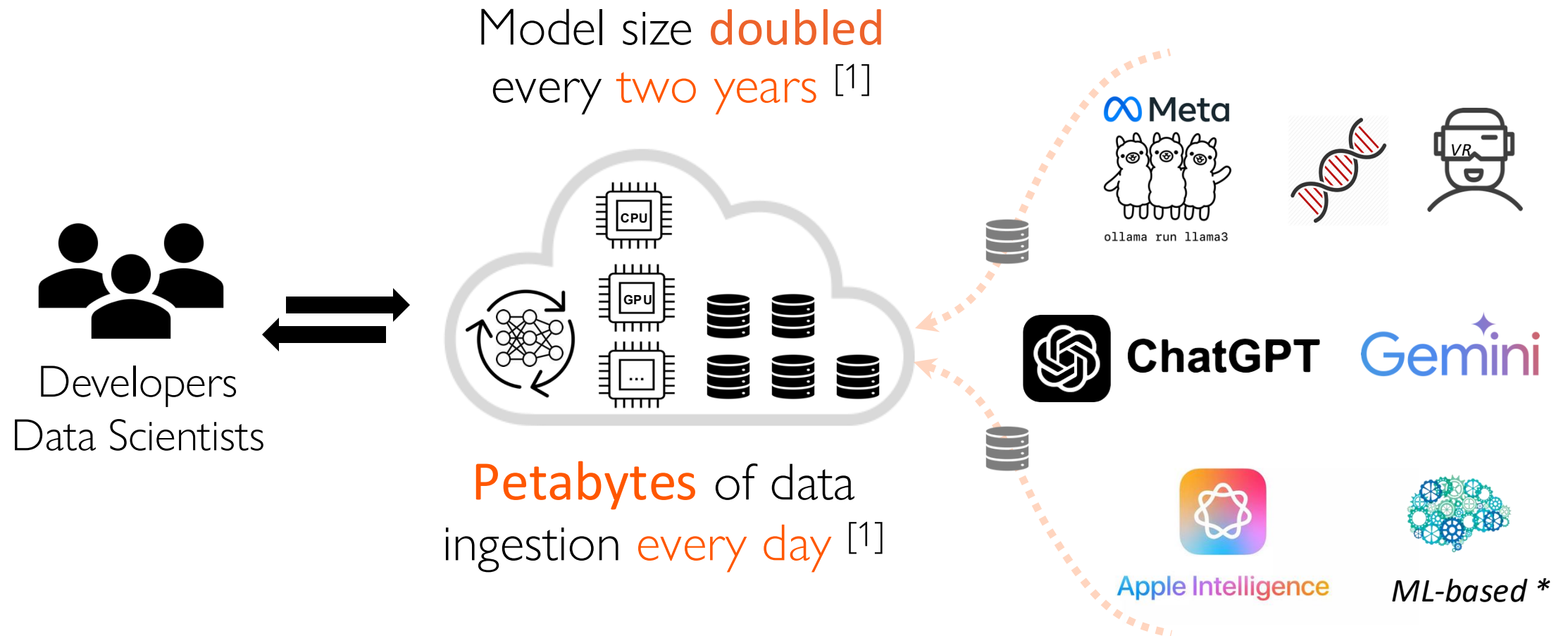


Cloud Datacenter: Scale-Out Servers



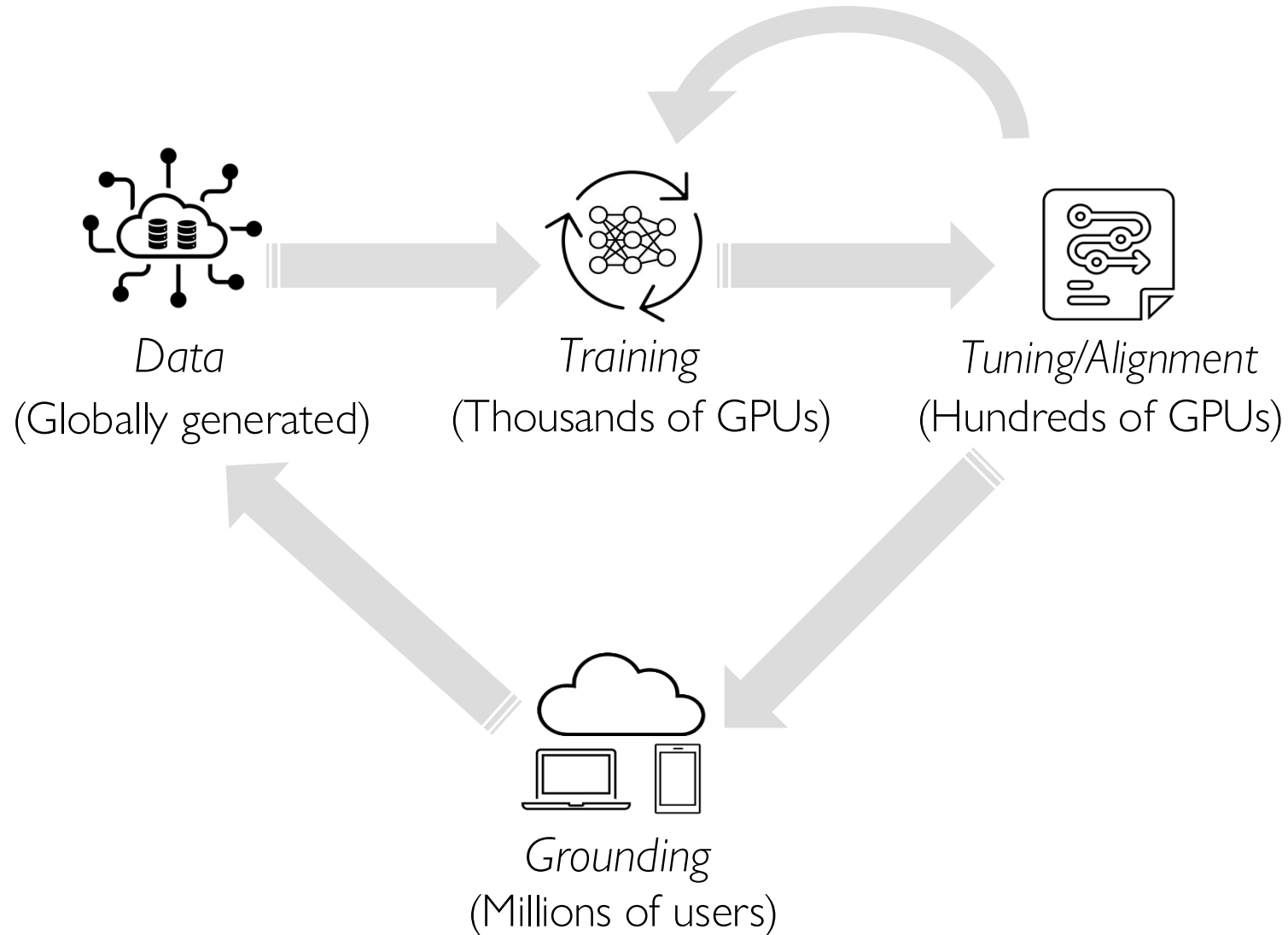
Meta's AI datacenter houses 16K of its 24K GPUs for training the latest Llama 3 model

Cloud Faces Skyrocketing ML Workloads

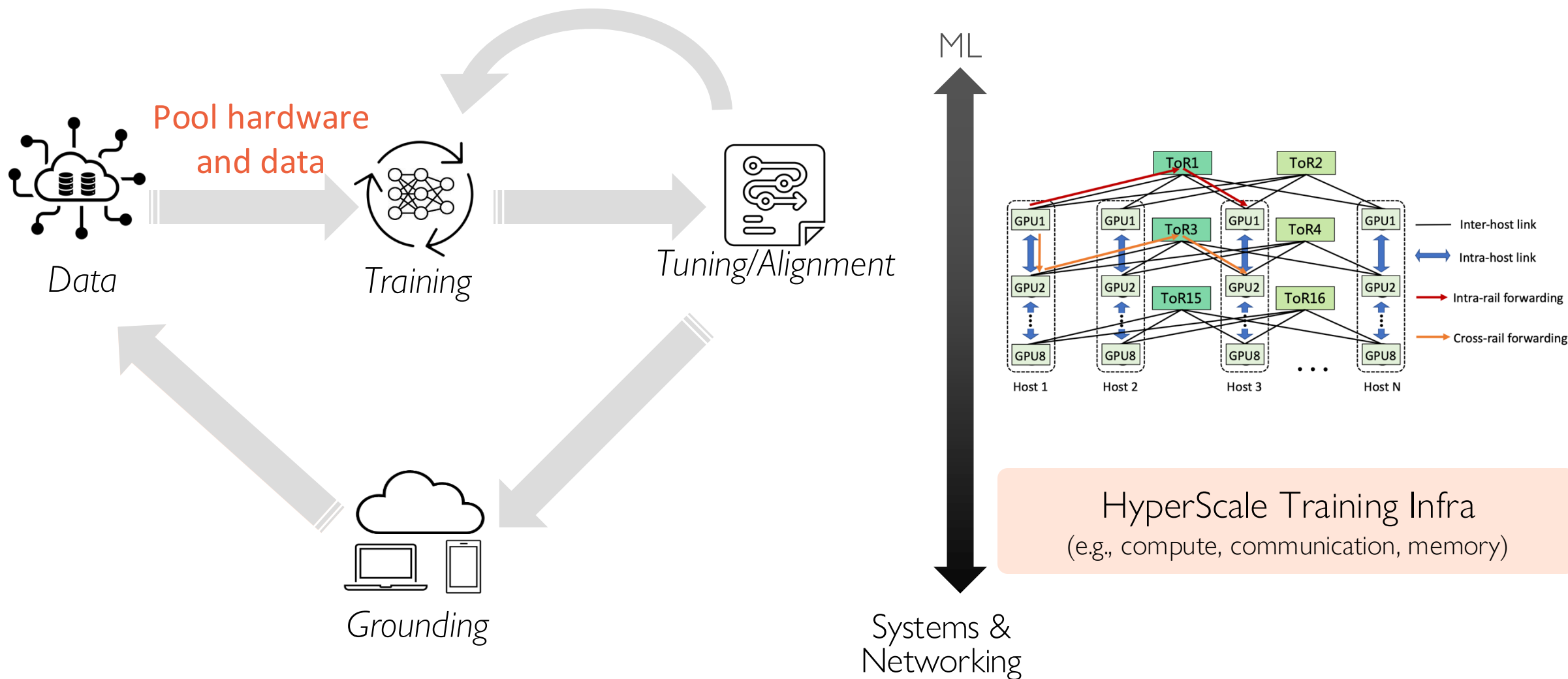


[1] "Sustainable AI: Environment Implications, Challenges and Opportunities", Meta, MLSys'22

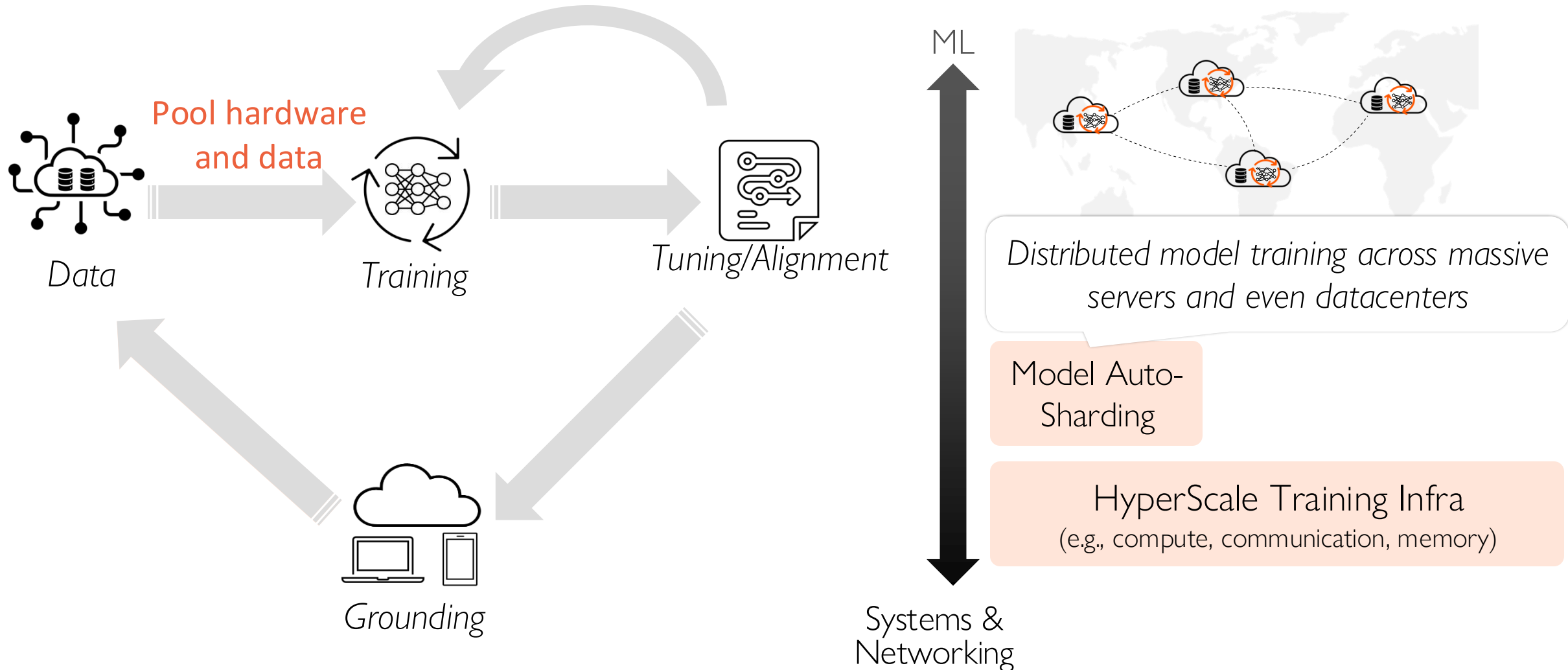
GenAI Lifecycle Demands Great Systems Support



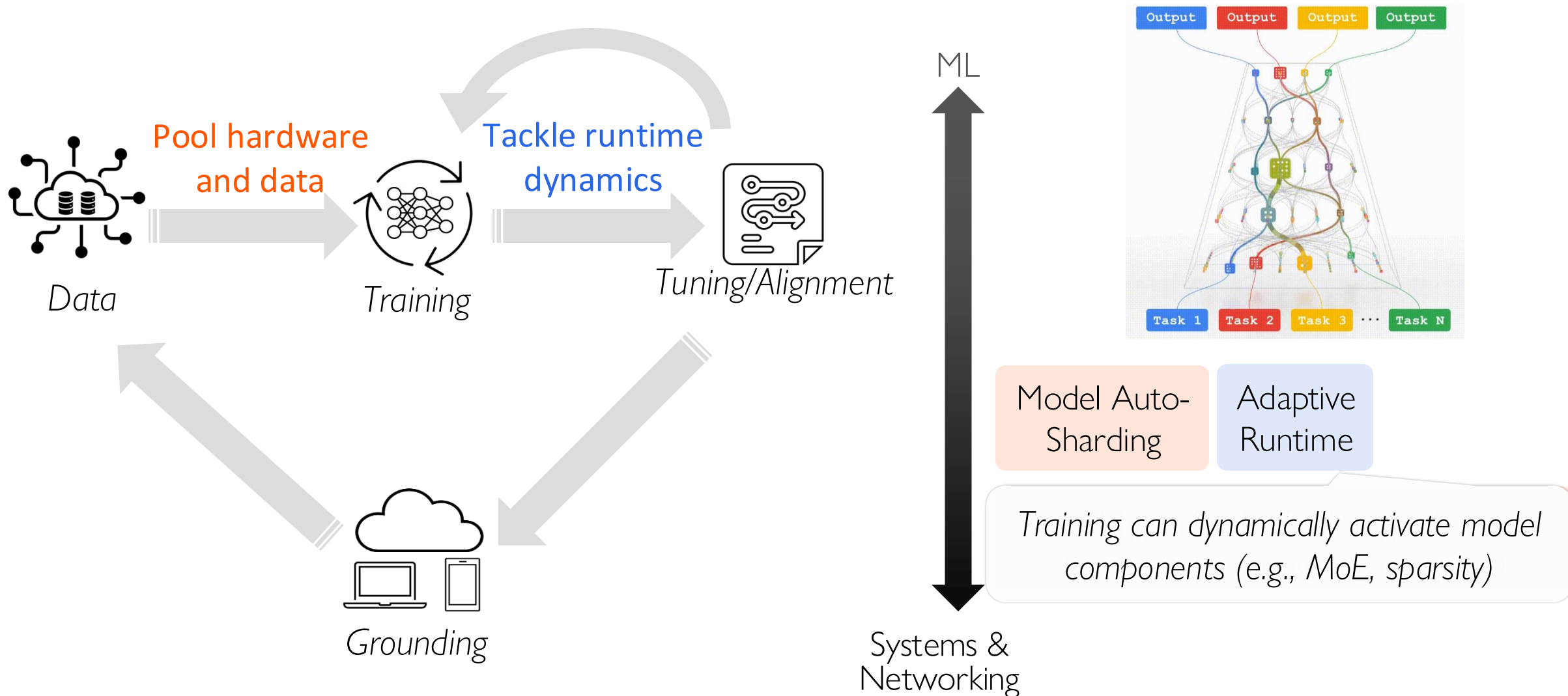
GenAI Lifecycle Demands Great Systems Support



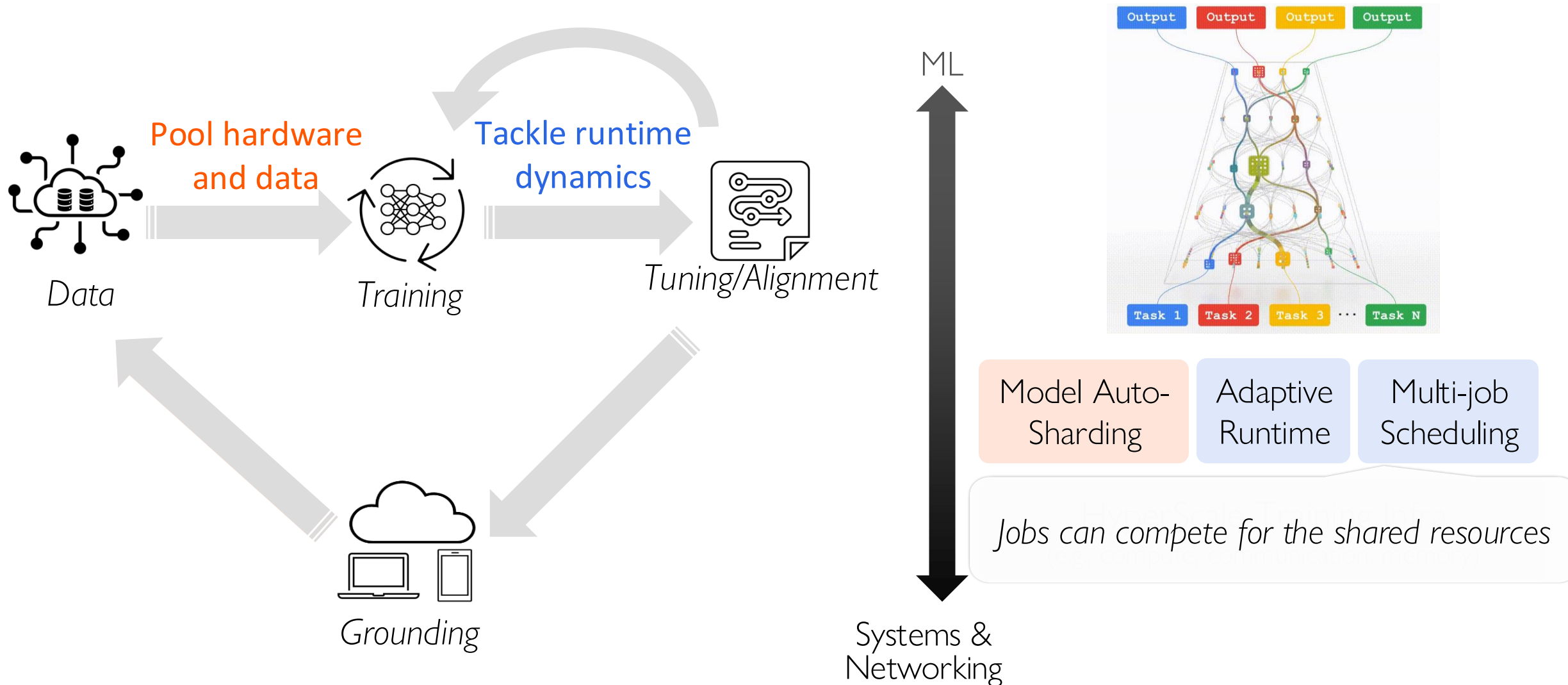
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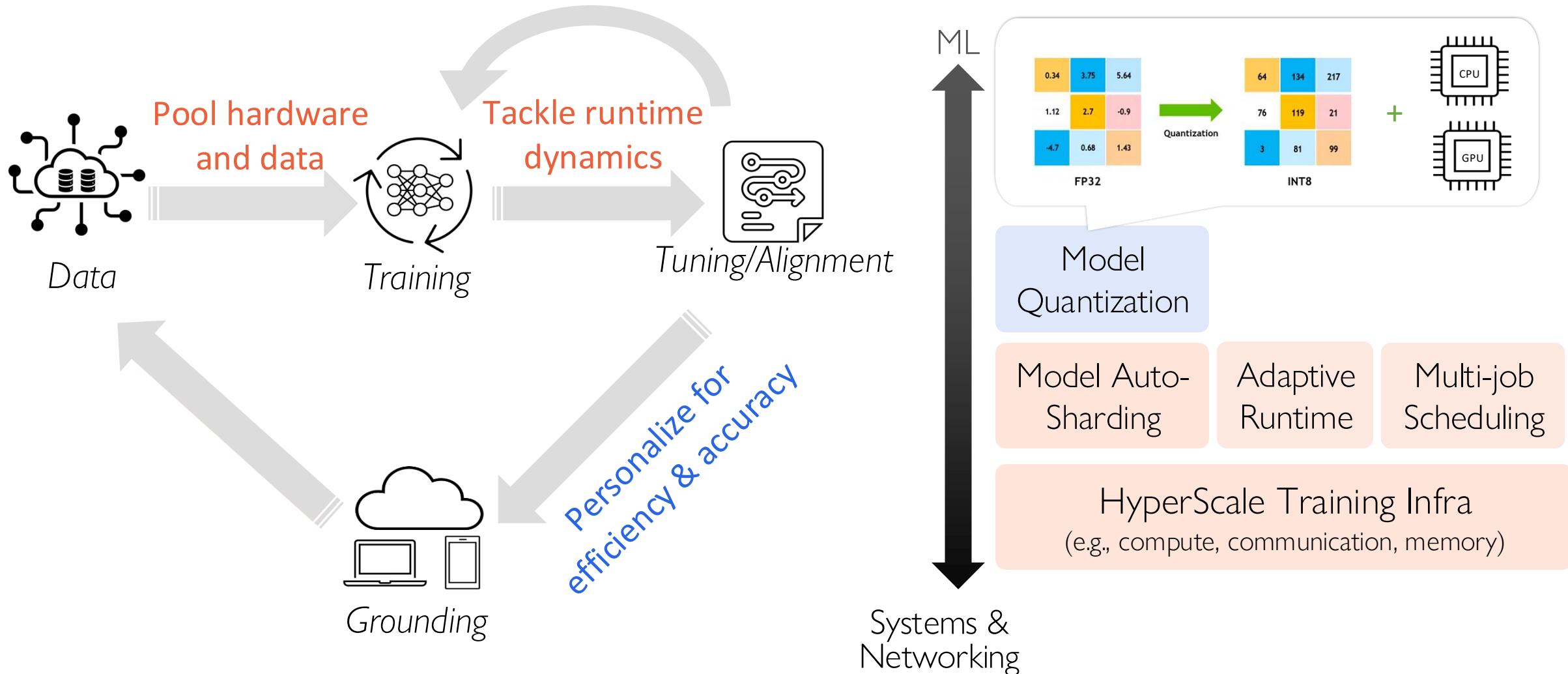
GenAI Lifecycle Demands Great Systems Support



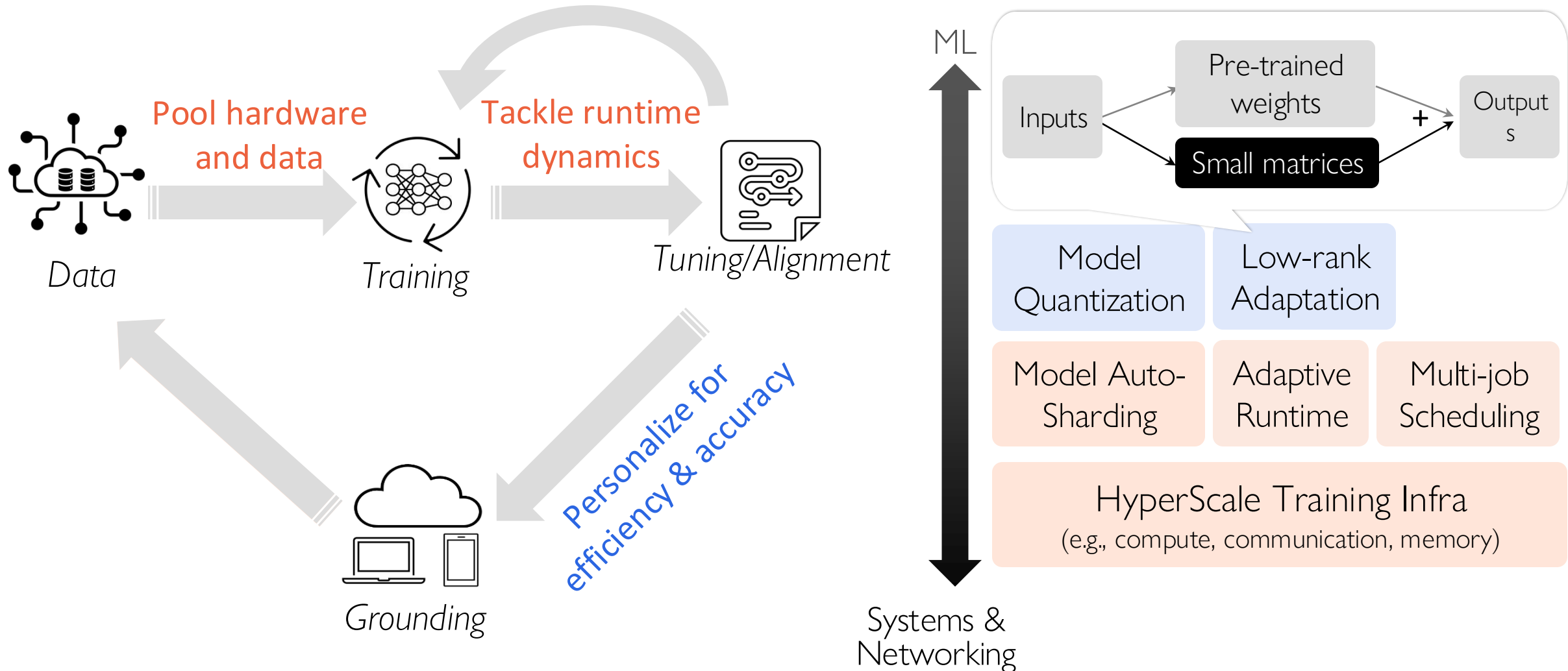
GenAI Lifecycle Demands Great Systems Support



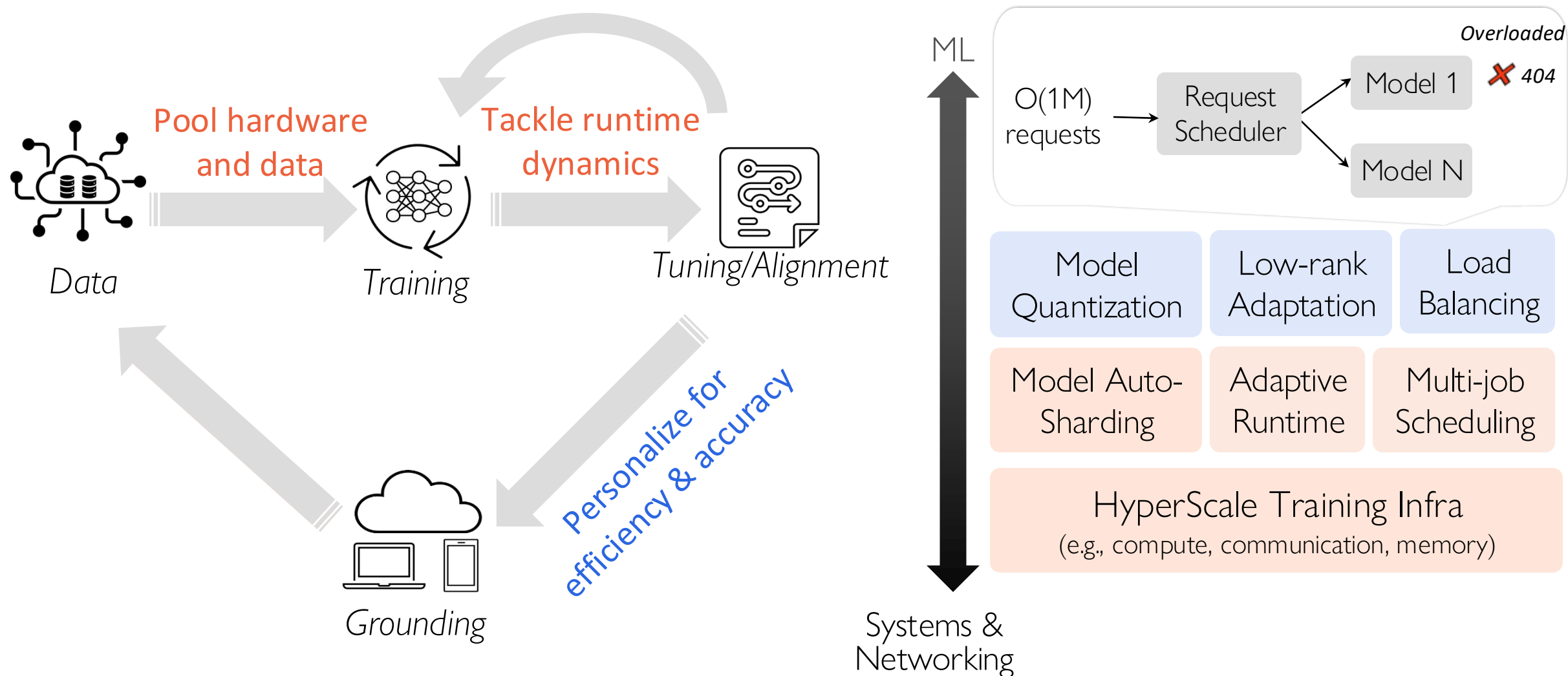
GenAI Lifecycle Demands Great Systems Support



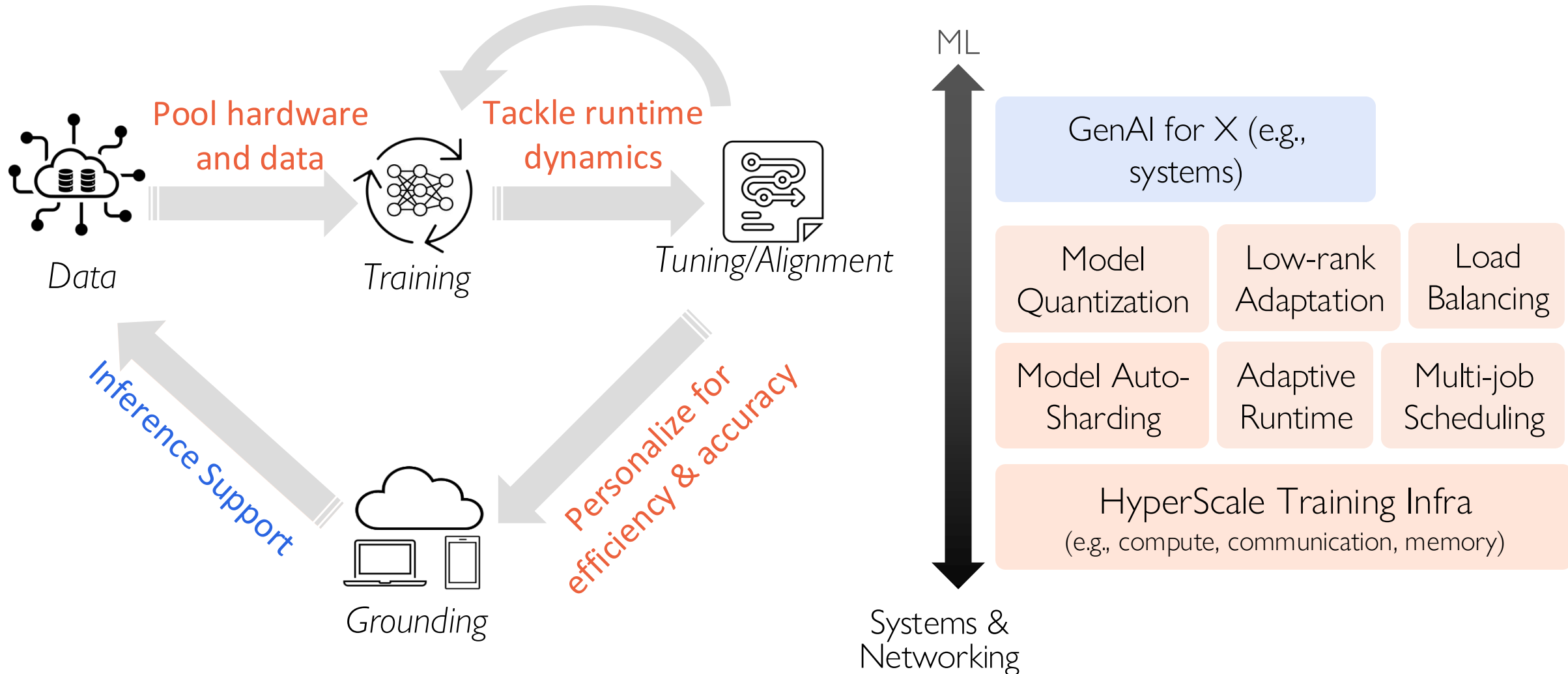
GenAI Lifecycle Demands Great Systems Support



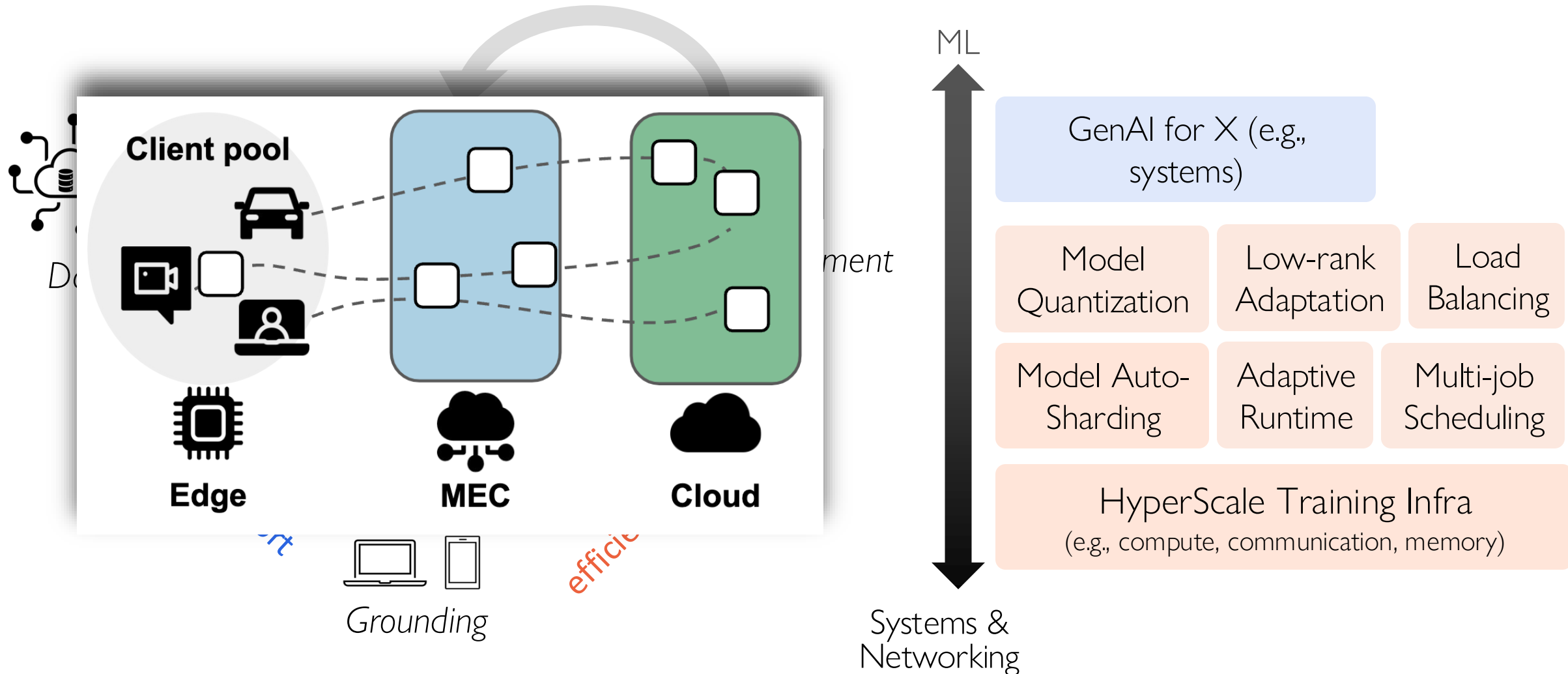
GenAI Lifecycle Demands Great Systems Support



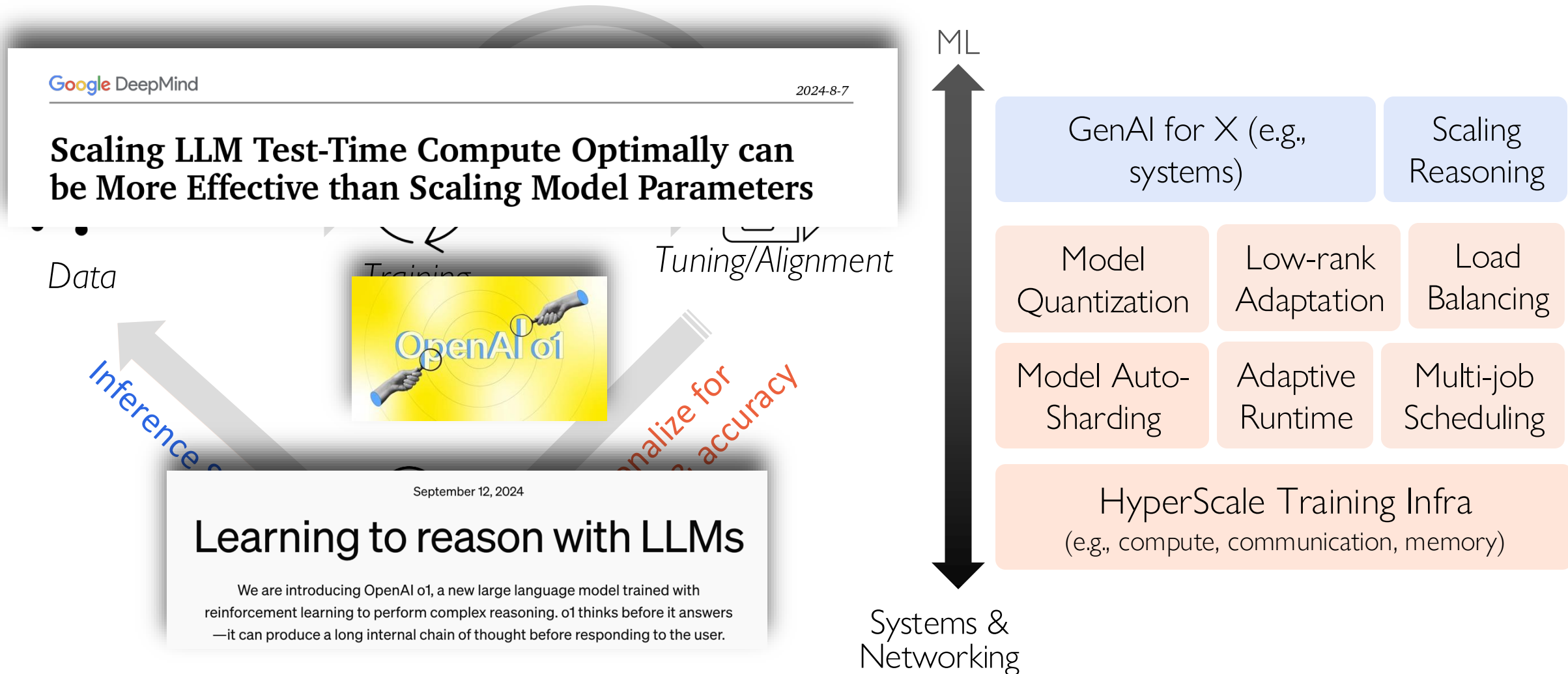
GenAI Lifecycle Demands Great Systems Support



GenAI Lifecycle Demands Great Systems Support



GenAI Lifecycle Demands Great Systems Support



What will you learn in this course?

-- How to **Support Practical ML**

Efficiency: Efficient use of hardware for high scalability and throughput

Effectiveness: High accuracy and fast convergence

Easy to use: Improve development productivity for developers and users

Without ML Systems



Researcher

ResNet
Transformer

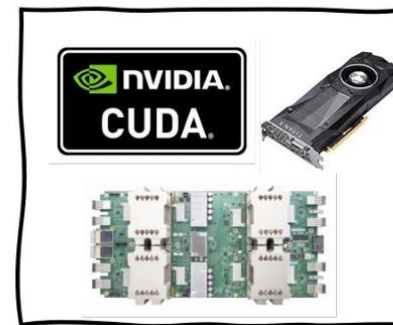
ML Research

44k lines of code

Six months



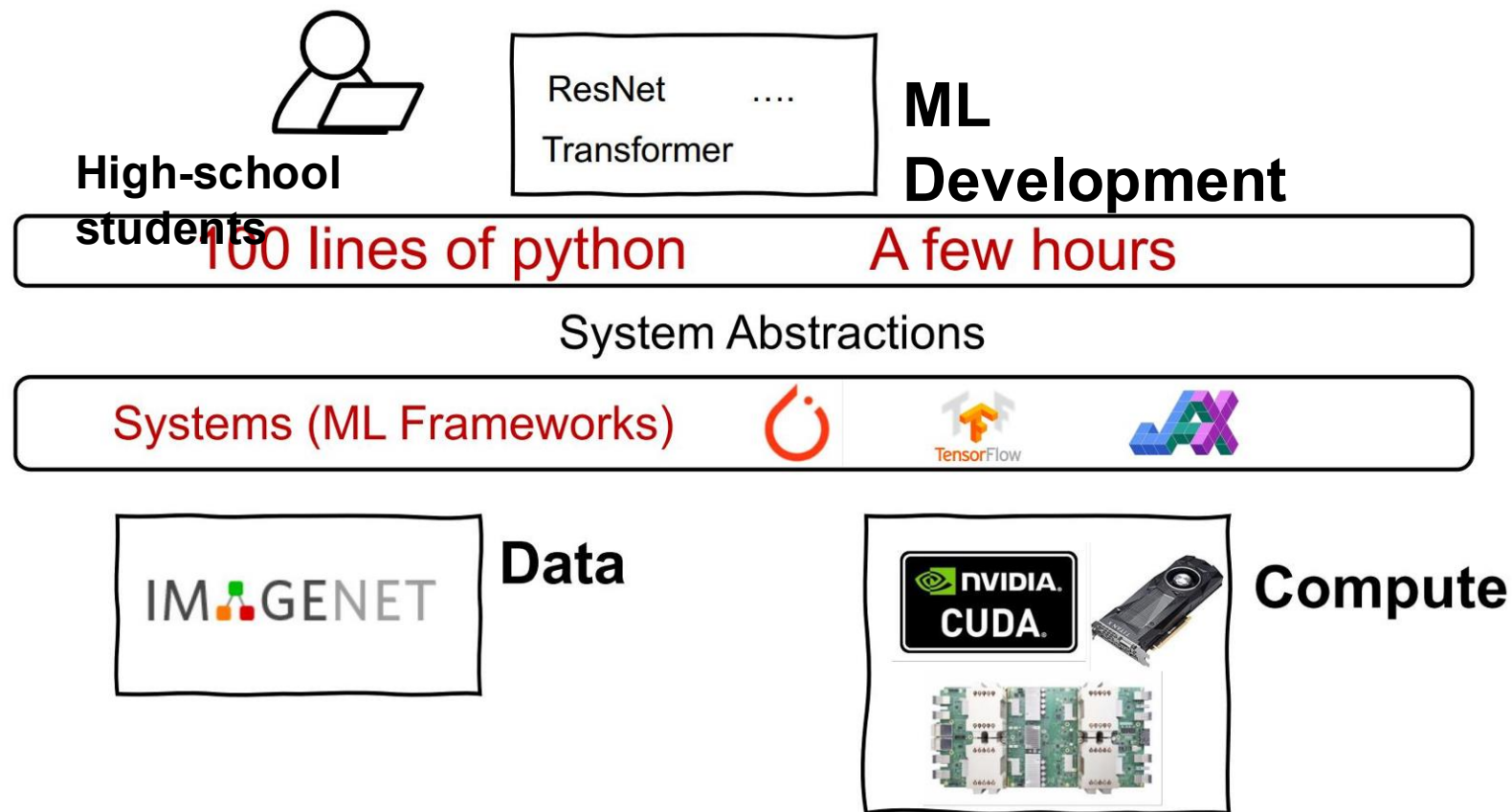
Data



Compute

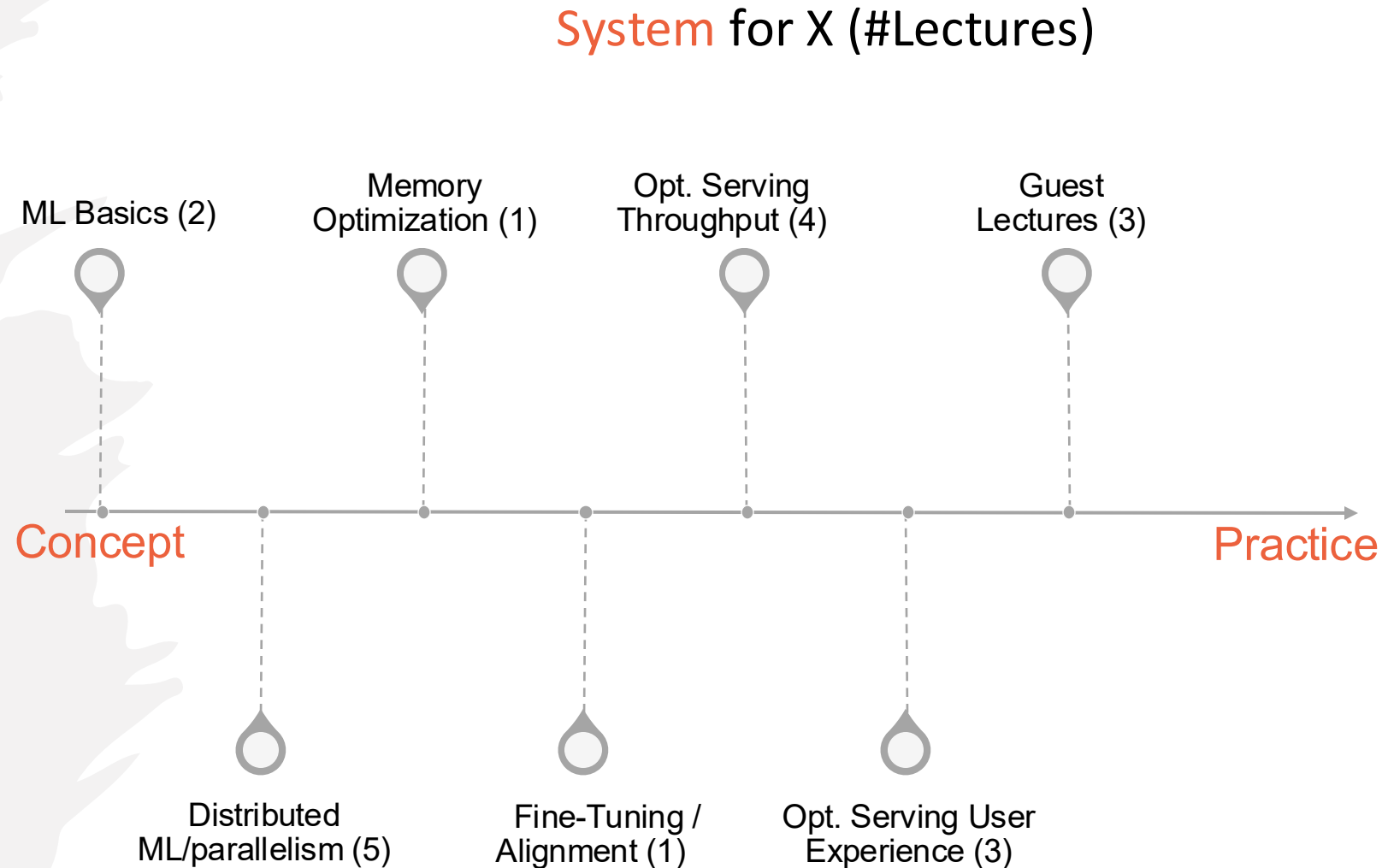
As ML Systems Democratize

AI



What will you learn in this course?

-- How to **Support Practical ML**



What will you learn in this course?

-- How to **Support Practical ML**

Learning Outcomes

After completing this course you should be able to

- Understand the basic functioning of ML Sys
- Articulate and critique latest GenAI Sys
- Be prepared to perform new research

This is a **Seminar-like** Course.
The more **active** the better.

Have Fun!

Agenda

- What will you learn in this course?
 - GenAI, systems, networks, ...
- What will you do in this course?
 - Participation, assignment, course project, ...

Course Expectations (Grading)

	Weight	Notes
Participation	10%	Up to two absences

While we value teamwork, clarify **each member's** contribution in group **report**.

Grading Scheme (grade is the **better** of the two)

Grade	Absolute Cutoff ($\geq$)	Relative Bin ($\geq$)
A+	95	Top 5%
$\geq$ A-	85	Top 35%
$\geq$ B-	70	Top 80%
$\geq$ C-	55	Hard to Believe!
$<$ C-		Unbelievable!!

Out of 150+ students, only 3 got $\leq$ B in my previous course. 😊

- Use Canvas to find potential group members

- Group size should be 4-5

Form Groups ASAP • May allow a few smaller groups if/when students drop out

- Submit at <https://forms.gle/TVkqSQAvUfKzsZGA7>
 - By Sept 8th the latest, but **right now** is better

Course Expectations (Grading)

	Weight	Notes
Participation	10%	Up to two absences
(Group) Panel Discussion	6%	Two discussions (3%+3%)

Participation & Paper Summaries

- Up to **two absences** in both attendance and paper summaries
 - Write summaries for **8 out of 10** papers (feel free to pick 8; 3% each)

Date	Topic	Lecturer	Slides	Assignment/Discussion
Aug 27	Course Introduction and Logistics	Fan Lai		
Aug 29	Transformers	Fan Lai		
Sept 3	Transformers Deep Dive	Fan Lai		
Sept 5	Distributed Training Overview	Fan Lai		DeepSeek-V3 Report (Sec 1-3)
Sept 10	Data Parallelism	Fan Lai		
Sept 12	Tensor Parallelism	Fan Lai		LlamaRL

First summary
due

Please upload your PDF summary to Canvas by 11:59 PM **before** lecture day.

Participation & Paper Summaries

- Each summary (4-5 **paragraphs**) must follow the template
 - What is the problem and why is it important?
 - The main proposed idea(s).
 - A summary of your understanding of different components of the proposed technique, e.g., the purpose of critical design choices.
 - What are the **technical** drawbacks/limitations
 - How will you improve them?

Please upload your PDF summary to Canvas by 11:59 PM **before** lecture day.

Participation & Paper Summaries

- Each summary (4-5 **paragraphs**) must follow the template
 - What is the problem and why is it important?
 - The main proposed idea(s).
 - A summary of your understanding of different components of the proposed technique, e.g., the purpose of critical design choices.
 - What are the **technical** drawbacks/limitations
 - How will you improve them?

Grading criteria, each summary has 10 points in total. For each item here, you get:

- 2: The summary item demonstrates a clear understanding of the paper.
- 1: The summary item misses the point of the paper.
- 0: The summary item is missing.

Lab Assignments

- Two Assignments (10% each)
 - Likely will use NCSA clusters for GPUs; Remember to *start early* due to GPU access
 - We will provide details later
- Topics
 - Distributed model training
 - Inference optimizations

Panel Discussion (After Normal Lecture)

- The Authors (assigned)
 - ‘Companion’ group that **answers** questions from ‘Reviewers’
- The Reviewers (assigned)
 - ‘Reviewer’ group that **poses** challenging **research** questions
- Rest of the Class & **Instructor**
 - Ask questions directly too

Each group will be assigned to these slots at least once; 2 discussions per group (only! 😊)

Summary of Your Role

Submit Paper Summary
(if assigned)

Be Engaged During
Lecture

One day
before lecture

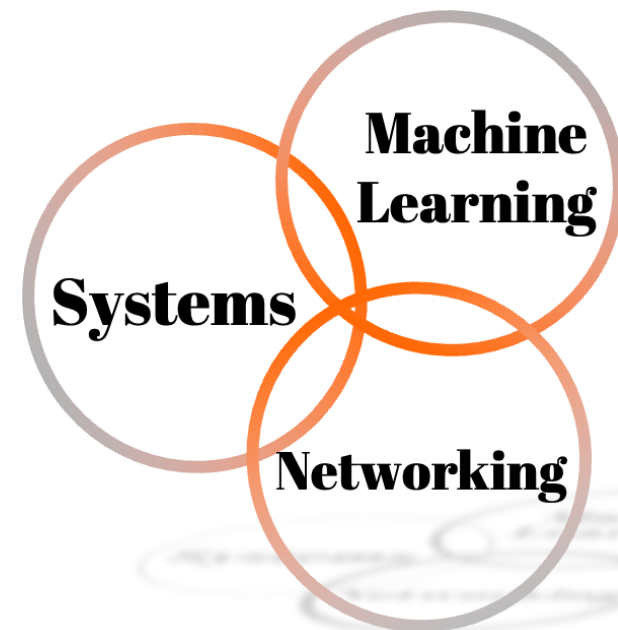
Right after
teaching

Post-
lecture

Attend the Lecture
(Up to 2 absence)

~20 min Pannel Discussion
(if there is paper
summary)

Systems for ML Projects



Research-Oriented Course Project

- The final project accounts for **40%** of total grades
- What can and cannot be a project?
 - Just surveys are not allowed
 - Measurements of new environments or of existing solutions in new environments are acceptable
 - Proposing new research problems or addressing existing ones is **highly preferred!**
- An ideal project should answer the questions you asked during paper summary and points you cared about in panel discussion

How to Approach it?

1. Find a problem and motivate why this is worth solving
2. Quickly survey background and related work
 - Might require you to go back to the first step
3. Form/update your hypothesis
4. Test your hypothesis
 - Go back to 3 until you are happy
5. Present your findings to the instructor and in writing
 - Discuss known limitations

Draft Proposal (Oct 1st; 5%)

- Two pages ([template](#)), plus references, **must** include
 - *Problem*: What is the problem?
 - *Motivation*: Why is it important?
 - *Solution overview*: Any initial thoughts?
 - *Evaluation plan*: How would you evaluate it?
- Submit as a group, including your group information (*form a group ASAP*)
- Approved by the instructor and agreed upon by you
 - Forms the basis of expectation

Mid-Semester Report (Nov 7; 5%)

- Dedicated proposal feedback session (No class on Oct 3, 8, 10)
 - This is to make sure you are receiving feedback and making progress
 - Feel free to stop by *office hours on any day!*
- Extend your proposal to 3-4 pages ([template](#)), which must include
 - What is the problem?
 - Why is it important? (Motivating experiments)
 - What are the most related work?
 - What's your hypothesis so far?
 - How are/will you evaluate it?

Final Presentation (Dec 3, 5, 10; 15%)

- In-class (group) presentation
 - Please spend a significant amount of time on working your project and making this presentation nice and clear.
- Graded by instruction team (50%) and your classmates (50%)
 - **Instructor:** based on format, correctness, depth, clarity, insights
 - **Peers:** make sure your classmates feel they indeed learn something after listening to your presentation

Final Report (Dec 19; 25%)

- Final report (7-8 pages excluding references by Dec 19):
 - Should be written like the papers you've read
 - Abstract, Introduction
 - Background/Motivation
 - Design
 - Implementation and Evaluation
 - Conclusion
 - As if you'd submit it to a workshop with ~3 more months of work or to a conference after ~6 more months of work
 - [How to Write a Great Research Paper](#) by Simon Peyton Jones

Project Ideas

- Some potential project ideas (let's chat more)
 - Multi-tier memory, including CXL and RDMA, for long-context LLMs
 - Energy-efficient GenAI serving on end-user devices
 - Systems support for reinforcement learning from human feedback
 - Making chatbots real by allowing them to cut in
 - Repurposing spatiotemporal redundancy in video generation
 - Benchmark important DL workloads to identify important angles
 - ML for systems optimizations (e.g., networking problems)
 - ...
- Any of your crazy ideas!

Thank you!

Date	Milestone	Details
09/08	Form Group	Find 4-5 like-minded teammates
10/01	Submit Proposal	Send your proposal by email to receive feedback either via email or in-person or both
10/03, 10/08, 10/10	Proposal Feedback (no class)	Refine the problem, overview, and form initial hypothesis/idea
11/01	Mid-term Report	3/4-page report
12/03 12/05 12/10	In-Class Project Presentations	Present your research findings
12/19	Research paper	Submit a report like the papers you read

NO extensions! Delayed submission will receive NO CREDIT

Thank you!

Attendance & Summary

- Up to 2 absences
- 8 out of 10 papers

Pannel Discussion

- Companion + Reviewer

Form Groups of 4-5; [submit](#) by *Sept 8*

Two Lab Assignments

- Distributed training
- Inference optimizations

Semester-long Project

- Final presentation
- Final report